

```
92 • DESCRIBE students;
93 • DESCRIBE books;
94 • DESCRIBE borrow_record;
95 • select * from students;
```

Result Grid



Filter Rows:

Export:



Wrap Cell Content:

	Field	Type	Null	Key	Default	Extra
►	id	int	NO	PRI	NULL	auto_increment
	students_name	varchar(45)	NO		NULL	
	age	int	NO		NULL	
	gender	char(2)	YES		NULL	
	telephone	varchar(11)	YES		NULL	

```
92 • DESCRIBE students;
93 • DESCRIBE books;
94 • DESCRIBE borrow_record;
95 • select * from students;
```

Result Grid



Filter Rows:

Export:



Wrap Cell Content

	Field	Type	Null	Key	Default	Extra
▶	book_id	int	NO	PRI	NULL	auto_increment
	book_name	varchar(50)	NO		NULL	
	author	varchar(30)	YES		NULL	
	category	varchar(20)	YES		NULL	

91

92 • DESCRIBE students;

93 • DESCRIBE books;

94 • DESCRIBE borrow_record;

95 • select * from students;

Result Grid



Filter Rows:

Export:



Wrap Cells

	Field	Type	Null	Key	Default	Extra
▶	borrow_id	int	NO	PRI	NULL	auto_increment
	student_id	int	NO	MUL	NULL	
	book_id	int	NO	MUL	NULL	
	borrow_date	date	NO		NULL	
	return_date	date	YES		NULL	

```
95 • select * from students;
96 • select * from books;
```

Result Grid



Filter Rows:

Edit:



	id	students_name	age	gender	telephone
▶	1	张三	20	男	13800138001
	2	李四	19	女	13800138002
	3	王五	21	男	13800138003
	4	赵六	19	女	13800138004
	5	孙七	22	男	13800138005
	6	周八	23	女	13800138006
	7	吴九	20	男	13800138007
	8	郑十	19	女	13800138008
	9	钱一	22	男	13800138009
	10	冯二	21	女	13800138010
	11	陈三	18	男	13800138011
	12	褚四	23	女	13800138012
	13	Lio	19	男	13800138013
●	NULL	NULL	NULL	NULL	NULL

Result 1

Result 2

Result 3

students 4 ×

books 5

b

6 • `select * from books;`

Result Grid



Filter Rows:

Edit:



book_id	book_name	author	category
1	西游记	吴承恩	古典文学
2	西游记	吴承恩	古典文学
3	三国演义	罗贯中	古典文学
4	红楼梦	曹雪芹	古典文学
5	三体	刘慈欣	科幻小说
6	三体	刘慈欣	科幻小说
7	Python编程	张三峰	计算机
8	Java入门	李四光	计算机
9	数据库原理	王五明	计算机
10	平凡的世界	路遥	现代文学
11	围城	钱钟书	现代文学
12	解忧杂货店	东野...	外国文学
NULL	NULL	NULL	NULL

Result 1

Result 2

Result 3

students 4

books 5 ×

97 • `select * from borrow_record;`

98

qq -- 1 查询没有借阅《西游记》的所有学生的姓名和年

Result Grid |   Filter Rows: | Edit:    | Export/Import: 

borrow_id	student_id	book_id	borrow_date	return_date
1	1	1	2026-01-01	NULL
2	1	3	2026-01-02	2026-02-10
3	2	2	2026-01-03	NULL
4	2	5	2026-01-05	2026-03-01
5	3	6	2026-01-08	NULL
6	3	7	2026-01-10	NULL
7	4	4	2026-01-12	2026-02-20
8	4	8	2026-01-15	NULL
9	5	1	2026-01-18	NULL
10	5	9	2026-01-20	2026-03-05
11	6	2	2026-02-01	NULL
12	6	10	2026-02-03	2026-03-10
13	7	5	2026-02-05	NULL
14	7	11	2026-02-08	NULL
15	8	3	2026-02-10	2026-03-15

Result 1 Result 2 Result 3 students 4 books 5 borrow_record 6 ×

```
99      -- 1. 查询没有借阅《西游记》的所有学生的姓名和年龄
100 •   select students_name,age from students s
101     where not exists (
102         select id from borrow_record br
103         join books b on b.book_id = br.book_id
104         where b.book_name = '西游记'
105         and s.id = br.student_id
106     );
```

Result Grid |   Filter Rows: | Export:  | Wrap Cell Co

	students_name	age
▶	王五	21
	赵六	19
	吴九	20
	郑十	19
	钱一	22
	冯二	21
	陈三	18
	褚四	23
	Lio	19


```
107 -- 2.查询借阅了书名为“西游记”的书籍的所有学生借阅的所有书籍名
108 • select distinct book_name from books b
109 join borrow_record br on b.book_id = br.book_id
110 where br.student_id in (
111     select distinct student_id
112     from borrow_record br2
113     join books b2 on br2.book_id = b2.book_id
114     where b2.book_name = '西游记'
115 );
116
```

Result Grid		Filter Rows:	Export:	Wrap Cell Content:
	book_name			
▶	西游记			
	三国演义			
	数据库原理			
	三体			
	平凡的世界			


```
117      -- 3. 查询至少借阅了两本书的学生的姓名和性别
118  •   select  students_name,gender from students s
119  where exists (
120      select student_id from borrow_record br
121      where s.id = br.student_id
122      group by br.student_id
123      having count(book_id)>1
124  );
```

Result Grid



Filter Rows:

Export:



Wrap Cell

	students_name	gender
▶	张三	男
	李四	女
	王五	男
	赵六	女
	孙七	男
	周八	女
	吴九	男
	钱一	男

```
126      -- 4. 查询姓名为“张三”的学生没有借阅的书籍
127  •   select distinct book_name from books
128  ⊖   where book_id not in (
129      select book_id from borrow_record br
130      join students s on s.id = br.student_id
131      where s.students_name = "张三"
132  );
133      -- 5. 查询借阅了姓名为“张三”的学生借阅的至少一本书
```

Result Grid



Filter Rows:

Export:



Wrap Ce

	book_name
▶	西游记
	红楼梦
	三体
	Python编程
	Java入门
	数据库原理
	平凡的世界
	围城
	解忧杂货店

```
133  -- 5.查询借阅了姓名为“张三”的学生借阅的至少一本书籍的学生的学号和姓名
134  •  select id,students_name from students
135  where id in (
136      select student_id from borrow_record
137      where book_id in (
138          select book_id from borrow_record br
139          join students s on s.id = br.student_id
140          where s.students_name = '张三')
141  );
```

Result Grid



Filter Rows:

Edit:



Export/Import:



Wrap

	id	students_name
▶	1	张三
	5	孙七
	8	郑十
●	NULL	NULL

-- 6. 查询和姓名为“陈三”的同学借阅书籍集合相同的学生姓名

```
select distinct s.students_name from students s
join borrow_record br1 on s.id = br1.student_id
where br1.book_id in (
    select book_id from borrow_record
    where student_id = (select id from students where students_name = "陈三")
)
and not exists (
    select 1 from borrow_record br2
    where br2.student_id = s.id
    and br2.book_id not in (
        select book_id from borrow_record
        where student_id = (select id from students where students_name = "陈三")
    )
);
```

Result Grid



Filter Rows:

Export:



Wrap Cell Content:



	students_name
冯二	
陈三	

```
158      -- 7. 检索所有学号比姓名为"占位符"的同学大，而年龄比ta小的学生姓名
159 •      select students_name from students
160      where id > (select id from students where students_name = "张三" )
161      and age < (select age from students where students_name = "张三");
162
```

Result Grid



Filter Rows:

Export:



Wrap Cell Content:



	students_name
▶	李四
	赵六
	郑十
	陈三
	Lio

```
163      -- 8. 检索姓名以L打头的所有学生的姓名和年龄
164  ●    SELECT students_name, age
165      FROM students
166      WHERE students_name LIKE 'L%';
```

Result Grid



Filter Rows:

Export:



Wri

	students_name	age
▶	Lio	19

```
168      -- 9. 求年龄大于女同学平均年龄的男同学的姓名和年龄
169      • select students_name,age from students
170      where gender = "男"
171      and age > (select avg(age) from students
172                  where gender = "女"
173                  group by gender
174                );
```

Result Grid |  Filter Rows: | Export:  | Wrap Cell Con

	students_name	age
▶	王五	21
	孙七	22
	钱一	22


```
175      -- 10. 求年龄小于所有女同学年龄的男同学的姓名和年龄
176  •   select students_name,age from students
177      where gender = "男"
178      and age < (select min(age) from students
179                  where gender = "女"
180      );
```

Result Grid |   Filter Rows: | Export:  | Wrap Cell Con

	students_name	age
▶	陈三	18

```
182      -- 3 . 1. 在学生表中插入一条已有学生id的学生信息，观察执行结果
183  • insert into students(id, students_name, age)
184    values (1, '重复ID学生', 20);
185
186      -- 2. 在学生表中插入一条年龄为10的学生信息，观察执行结果
```

Output



Action Output

	#	Time	Action	Message
✓	17	20:03:54	select students_name,age from students where gender = "男" and age > (select av...	3 row(s) returned
✓	18	20:04:10	select students_name,age from students where gender = "男" and age < (select mi...	1 row(s) returned
✗	19	20:04:28	insert into students(id, students_name, age) values (1, '重复ID学生', 20)	Error Code: 1062. Duplicate entry '1' for key 'students.PRIMARY'

```
186      -- 2. 在学生表中插入一条年龄为10的学生信息，观察执行结果
187 •    INSERT INTO students(id, students_name, age)
188      VALUES (10, '小学生', 10);
189
190      -- 3. 在借阅表中将某条记录的学生id修改为学生表中不存在的学生id，观察执行结果
```

Output



Action Output



	#	Time	Action	Message
✖	19	20:04:28	insert into students(id, students_name, age) values (1, '重复ID学生', 20)	Error Code: 1062. Duplicate entry '1' for key 'students.PRIMARY'
✖	20	20:05:17	INSERT INTO students(id, students_name, age) VALUES (10, '小学生', 10)	Error Code: 3819. Check constraint 'students_chk_1' is violated.

```
190  -- 3. 在借阅表中将某条记录的学生id修改为学生表中不存在的学生id, 观察执行结果
191  • UPDATE borrow_record
192  SET student_id = 999
193  WHERE student_id = 1;
194
195  -- 4. 基于“参照完整性约束”的知识, 针对借阅表中的两个外键 (书籍id和学生id),
196  -- 提供两个不满足参照完整性的插入操作
```

Output

 Action Output ▼

	#	Time	Action	Message	Duration / Fetch
✖	20	20:05:17	INSERT INTO students(id, students_name, age) VALUES (10, '小学生', 10)	Error Code: 3819. Check constraint 'students_chk_1' is violated.	0.000 sec
✖	21	20:05:39	UPDATE borrow_record SET student_id = 999 WHERE student_id = 1	Error Code: 1452. Cannot add or update a child row: a foreign key constraint fails ('li...	0.015 sec

```
195 -- 4.基于“参照完整性约束”的知识，针对借阅表中的两个外键（书籍id和学生id），
196 -- 提供两个不满足参照完整性的插入操作
197 -- 学生表没有 id=888 的学生
198 • INSERT INTO borrow_record(student_id, book_id)
199 VALUES (888, 1);
200 -- 书籍表没有 book_id=1 的书
```

Output



Action Output

	#	Time	Action	Message
✖	22	20:05:59	INSERT INTO borrow_record(student_id, book_id) VALUES (888, 1)	Error Code: 1364. Field 'borrow_date' doesn't have a default value
✖	23	20:06:17	INSERT INTO borrow_record(student_id, book_id) VALUES (888, 1)	Error Code: 1364. Field 'borrow_date' doesn't have a default value

```
200      -- 书籍表没有 book_id=666 的书
201  •  INSERT INTO borrow_record(student_id, book_id)
202      VALUES (1, 666);
203
```

Output



Action Output



	#	Time	Action	Message
✖	23	20:06:17	INSERT INTO borrow_record(student_id, book_id) VALUES (888, 1)	Error Code: 1364. Field 'borrow_date' doesn't have a default value
✖	24	20:06:29	INSERT INTO borrow_record(student_id, book_id) VALUES (1, 666)	Error Code: 1364. Field 'borrow_date' doesn't have a default value

```

206  -- 使得向借阅表插入一条记录时，检查该学生当前借阅书籍数目，
207  -- 如果小于3，成功借出，否则拒绝执行
208  DELIMITER $$
209  ● create trigger borrow_check_trg before insert on borrow_record
210    for each row
211    begin
212      if (select count(*) from borrow_record br
213          where br.student_id = new.student_id and br.return_date is null ) >= 3
214          signal sqlstate '45000' set message_text = "拒绝执行";
215      end if;
216
217    end$$
218  DELIMITER ;
219  -- 成功插入
220  insert into borrow_record (student_id,book_id,borrow_date,return_date)
221    value(3,4,'2026-04-03',null);
222  -- 失败插入

```

Output



Action Output



	#	Time	Action	Message
✖	27	20:07:27	insert into borrow_record (student_id,book_id,borrow_date,return_data) value(3,4,'...	Error Code: 1054. Unl
✔	28	20:07:41	insert into borrow_record (student_id,book_id,borrow_date,return_date) value(3,4,'...	1 row(s) affected


```

208 DELIMITER $$
209 • create trigger borrow_check_trg before insert on borrow_record
210 for each row
211 begin
212     if (select count(*) from borrow_record br
213         where br.student_id = new.student_id and br.return_date is null ) >= 3 then
214         signal sqlstate '45000' set message_text = "拒绝执行";
215     end if;
216
217 end$$
218 DELIMITER ;
219 -- 成功插入
220 insert into borrow_record (student_id,book_id,borrow_date,return_date)
221 value(3,4,'2026-04-03',null);
222 -- 失败插入
223 • insert into borrow_record (student_id,book_id,borrow_date,return_date)
224 value(7,10,'2026-04-03',null);

```

Output



Action Output

	#	Time	Action	Message
✓	28	20:07:41	insert into borrow_record (student_id,book_id,borrow_date,return_date) value(3,4,'...	1 row(s) affected
✗	29	20:08:19	insert into borrow_record (student_id,book_id,borrow_date,return_date) value(7,10,...	Error Code: 1644. 拒绝执行

```
226 -- 5. 数据安全性
227 -- 1. 使用root创建用户teacher和student，赋予用户teacher对使用的数据库的所有权限
228 • create user 'teacher'@'localhost' identified by '123456';
229 • create user 'student'@'localhost' identified by '123456';
230 • grant all privileges on library_system.* to 'teacher'@'localhost' with grant option;
231 • flush privileges;
```

Output



Action Output

	#	Time	Action	Message
✓	31	20:09:14	create user 'student'@'localhost' identified by '123456'	0 row(s) affected
✓	32	20:09:14	grant all privileges on library_system.* to 'teacher'@'localhost' with grant option	0 row(s) affected
✓	33	20:09:14	flush privileges	0 row(s) affected

```
C:\Users\Administrator>mysql -u teacher -p
```

```
Enter password: *****
```

```
Welcome to the MySQL monitor.  Commands end with ; or \g.
```

```
Your MySQL connection id is 21
```

```
Server version: 8.0.45 MySQL Community Server - GPL
```

```
Copyright (c) 2000, 2026, Oracle and/or its affiliates.
```

```
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owners.
```

```
Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.
```

```
mysql> USE library_system;
```

```
Database changed
```

```
mysql> INSERT INTO books(book_id, book_name, author)  
-> VALUES (100, 'MySQL从入门到精通', '张三');
```

```
Query OK, 1 row affected (0.01 sec)
```

```
C:\Users\Administrator>mysql -u student -p
Enter password: *****
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 22
Server version: 8.0.45 MySQL Community Server - GPL

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owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> USE library;
ERROR 1044 (42000): Access denied for user 'student'@'localhost' to database 'library'
mysql> USE library_system;
ERROR 1044 (42000): Access denied for user 'student'@'localhost' to database 'library_system'
mysql> SELECT * FROM library_system.books;
ERROR 1142 (42000): SELECT command denied to user 'student'@'localhost' for table 'books'
```

```
C:\Users\Administrator>mysql -u teacher -p
```

```
Enter password: *****
```

```
Welcome to the MySQL monitor.  Commands end with ; or \g.
```

```
Your MySQL connection id is 25
```

```
Server version: 8.0.45 MySQL Community Server - GPL
```

```
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```

```
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affiliates. Other names may be trademarks of their respective  
owners.
```

```
Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.
```

```
mysql> USE library_system;
```

```
Database changed
```

```
mysql> GRANT SELECT ON library_system.books TO 'student'@'localhost';
```

```
Query OK, 0 rows affected (0.01 sec)
```



Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

```
mysql> USE library;
```

```
ERROR 1044 (42000): Access denied for user 'student'@'localhost' to database 'library'
```

```
mysql> USE library_system;
```

```
ERROR 1044 (42000): Access denied for user 'student'@'localhost' to database 'library_system'
```

```
mysql> SELECT * FROM library_system.books;
```

```
ERROR 1142 (42000): SELECT command denied to user 'student'@'localhost' for table 'books'
```

```
mysql> SELECT * FROM library_system.books;
```

book_id	book_name	author	category
1	西游记	吴承恩	古典文学
2	西游记	吴承恩	古典文学
3	三国演义	罗贯中	古典文学
4	红楼梦	曹雪芹	古典文学
5	三体	刘慈欣	科幻小说
6	三体	刘慈欣	科幻小说
7	Python编程	张三峰	计算机
8	Java入门	李四光	计算机
9	数据库原理	王五明	计算机
10	平凡的世界	路遥	现代文学
11	围城	钱钟书	现代文学
12	解忧杂货店	东野圭吾	外国文学
100	MySQL从入门到精通	张三	NULL

13 rows in set (0.01 sec)

```
C:\Users\Administrator>mysql -u teacher -p
Enter password: *****
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 25
Server version: 8.0.45 MySQL Community Server - GPL

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affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> USE library_system;
Database changed
mysql> GRANT SELECT ON library_system.books TO 'student'@'localhost';
Query OK, 0 rows affected (0.01 sec)

mysql> REVOKE SELECT ON library_system.books FROM 'student'@'localhost';
Query OK, 0 rows affected (0.01 sec)
```



```
mysql> SELECT * FROM library_system.books;  
ERROR 1142 (42000): SELECT command denied to user 'student'@'localhost' for table 'books'  
mysql>
```

```
mysql> USE library_system;
Database changed
mysql>
mysql> -- 创建视图：只显示书名
mysql> CREATE VIEW book_name_view AS
    -> SELECT book_name FROM books;
Query OK, 0 rows affected (0.03 sec)

mysql> GRANT SELECT ON library.book_name_view TO 'student'@'localhost';
ERROR 1142 (42000): SELECT, GRANT command denied to user 'teacher'@'localhost' f
mysql> GRANT SELECT ON library_system.book_name_view TO 'student'@'localhost';
Query OK, 0 rows affected (0.01 sec)
```

```
mysql> SELECT * FROM library_system.book_name_view;
```

book_name
西游记
西游记
三国演义
红楼梦
三体
三体
Python编程
Java入门
数据库原理
平凡的世界
围城
解忧杂货店
MySQL从入门到精通

```
13 rows in set (0.00 sec)
```